Introduction to DNS

What is DNS?

- The "phonebook of the internet."
- Translates human-readable domain names (e.g., google.com) into IP addresses.

Why is DNS Important?

- Enables accessibility of websites.
- Supports distributed, scalable internet architecture.



Types of DNS Servers

1 Root Name Servers

Top-level servers handling .com, .org, etc.

2 TLD Name Servers

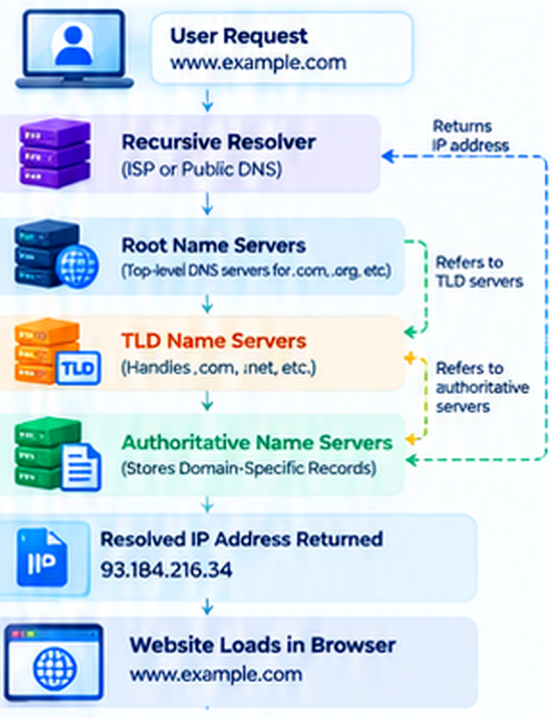
Specific to domains like .com, .net, etc.

3 Authoritative Name Servers

Store domain-specific records (e.g., IP addresses).

4 Recursive Resolvers

Handle queries on behalf of users, contacting multiple DNS servers if necessary.



DNS Caching and Performance Optimization



1 Why caching matters?

- Reduces latency.
- Reduces load on DNS servers.

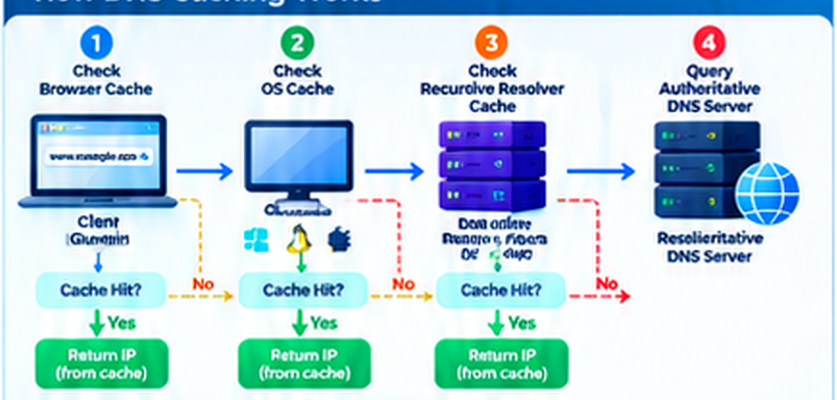
2 Where caching occurs?

- **Browser cache** Stores previous lookups.
- **OS cache** (e.g., hosts, Windows DNS cache).
- **Recursive Resolver cache** (ISP level).

3 Time-To-Live (TTL)

Determines how long a cached record is valid.

How DNS Caching Works



The Domain Name Resolution Process

1 User types a domain

You enter a domain name in the browser (e.g., www.example.com).

2 Browser cache check

Browser checks if the IP is already cached.

3 OS cache check

Operating system (OS) checks its local cache.

4 Query to local DNS resolver (ISP)

If not found, query goes to the configured DNS server (usually by your ISP).

5 Query to Root DNS servers

If still not found, the resolver queries a Root DNS server.

6 TLD name servers

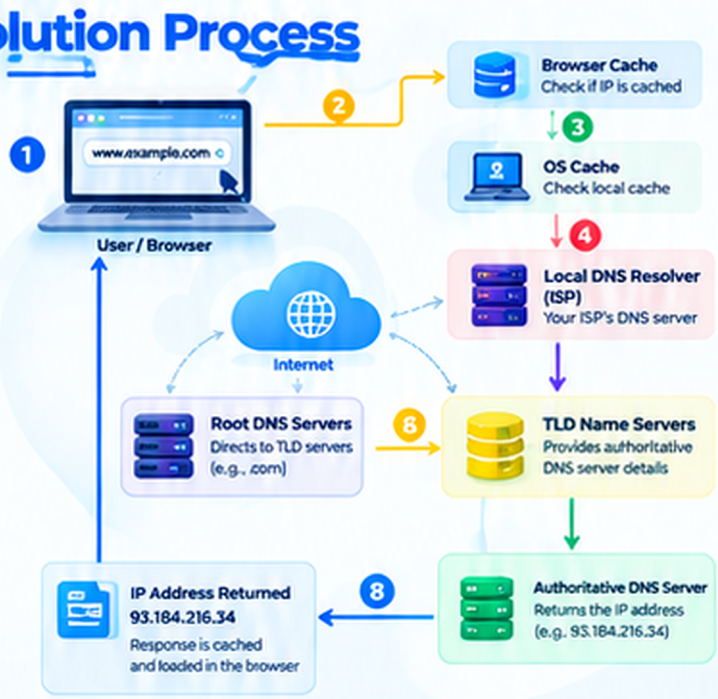
Root server refers to the TLD (e.g. .com) name servers, which provide the authoritative server details.

7 Authoritative DNS server

The authoritative DNS server for the domain returns the IP address.

8 IP returned & cached

The IP address is returned to the browser and cached for future use.



Importance of DNS in Large-Scale Systems

1 Ensuring High Availability

- Load balancing using DNS.
- Anycast DNS for faster global resolution.

3 DNS Failover Strategies

- Primary and secondary DNS servers.
- Automatic failover traffic during failures.

4 Content Delivery Networks (CDNs) & DNS

- How DNS directs users to the nearest CDN node for fast loading.
- Improves performance and reduces latency.

5 DNS Security Risks

- DNS poisoning, cache poisoning, and DDoS attacks.
- Use DNSSEC, rate limiting, and monitoring.



Interview Questions Related to DNS

1 Explain the DNS resolution process step by step.



2 What is the difference between authoritative and recursive DNS servers?



3 How does DNS caching improve performance? Where does it occur?



4 What is TTL in DNS, and why is it important?



5 How does DNS-based load balancing work in large-scale systems?



6 What are common DNS-related security threats, and how can they be mitigated?

